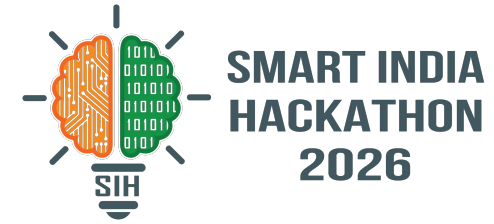


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26107
- **Problem Statement Title** - AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team ID** -
- **Team Name (Registered on portal)** - Apex

Problem : BIS compliance is complex and time-consuming, with standards, testing requirements, clauses, and certification information scattered across multiple sources. This makes it difficult for MSMEs, startups, and consumers to quickly identify the correct requirements and verify genuine BIS certification

PROPOSED SOLUTION

1. AI Assistant answers plain-English and voice queries.
2. One-Stop Engine delivers exact IS codes, clauses, and lab test lists instantly.
3. Interactive Roadmap provides a clear checklist from testing to license grant.
4. OCR Label Scanner verifies scanned rating plates against real format records.
5. Grounded BIS Retrieval anchors outputs strictly to verified official data.

Technology Stack

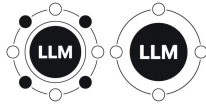
- **Frontend:** Next.js + React + Tailwind CSS



- **Backend:** FastAPI + Python



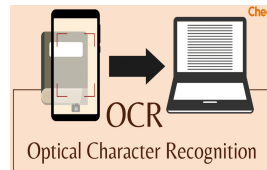
- **AI:** Gemini API / LLM



- **Database:** Verified BIS Knowledge Base



- **OCR:** Product label text extraction



- **API:** REST APIs



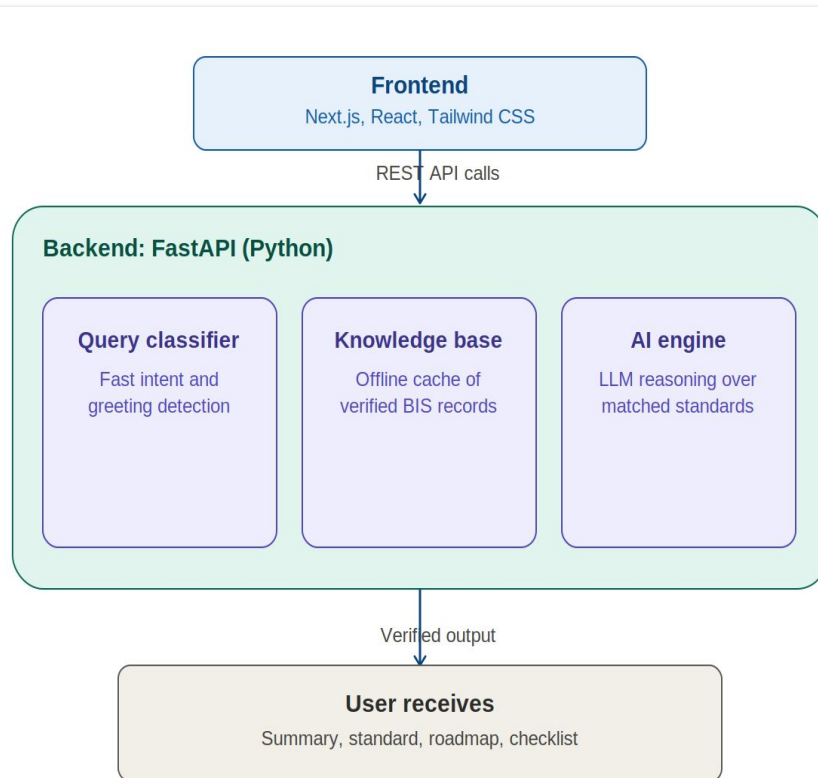
Core Modules

- AI Chat Assistant
- Standards & Evidence Engine
- Readiness Roadmap & Checklist
- OCR Label Scanner
- Product & Standards Catalog

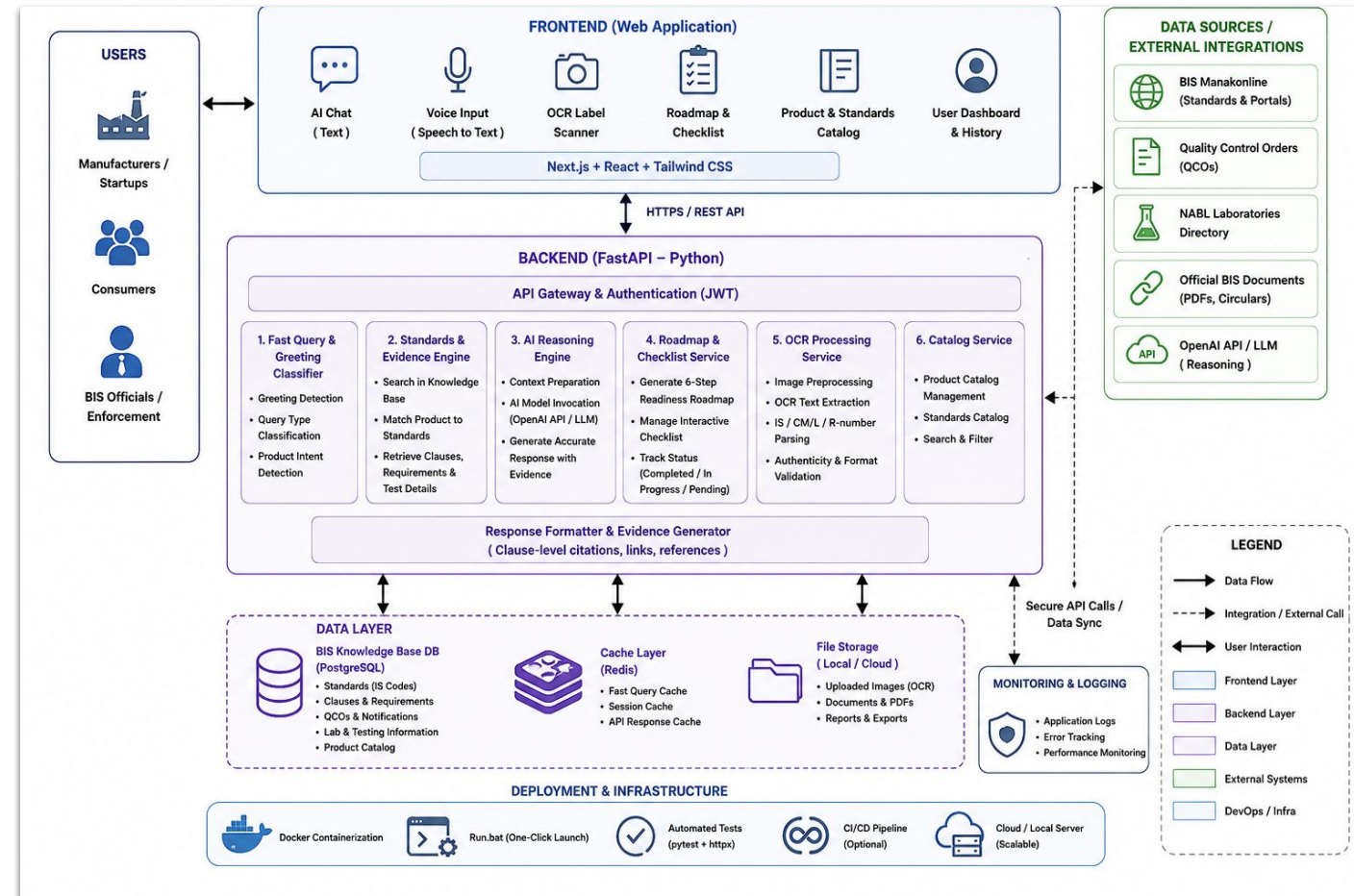
Implementation

- Fast local lookup for verified BIS records.
- AI used to understand queries and present results clearly.

Work Flow



Architecture Diagram



Why It Is Feasible?

- **Proven Tech Stack:** Python, FastAPI, React, OCR, LLMs.
- **Structured Data:** Offline-first knowledge base.
- **Web Architecture:** Easy deployment and scaling.
- **Modular Design:** Simple to add new standards.

Challenges & Risks

- **Dynamic Regulations:** Evolving BIS standards.
- **Image Quality:** Blurry scans reduce OCR accuracy.
- **Missing Data:** Unlisted products or standards.
- **API Access:** Requires official live portal integration.

Risk Mitigation

- **Evidence Grounding:** Pre-search BIS database.
- **Source Transparency:** Cite clauses and IS numbers.
- **Regular Updates:** Maintain versioned database.
- **Regex Validation:** Verify OCR-extracted numbers.
- **Automated Tests:** Built-in `pytest` execution.

Target Audience Impact:

- **MSMEs & Startups:** Reduces regulatory compliance research time from **weeks to seconds**, lowering consulting expenses.
- **Consumers:** Provides instant label verification to flag fake or substandard products.
- **BIS Officers:** Automates routine inquiries, freeing up time for field inspections and quality audits.

Economic & Social Benefits:

- **Economic:** Accelerates product time-to-market for domestic manufacturers, supporting **Make in India**.
- **Social:** Enhances public safety by curbing uncertified electrical and consumer goods

References and Research

- <https://www.bis.gov.in/>
- <https://www.crsbis.in/>
- <https://dpiit.gov.in/>
- <https://arxiv.org/abs/2005.11401>



BIS Sahayak

Compliance Assistant

NAVIGATION



AI Assistant



Scan & Verify



BIS Services



AI Readiness

Compliance profile

68

/ 100

Complete a scan to improve your business score.



BIS Sahayak AI

Authoritative Bureau of Indian Standards Decision Support • Powered by Gemini

API Online



Hello! I am **BIS Sahayak AI**, your authoritative compliance and standards assistant for the Bureau of Indian Standards.

Ask me about required standards, testing clauses, QCO mandates, or certification roadmaps for any product.

Listen

I want to manufacture an electric kettle

Which standard applies to a laptop?

What are the requirements for pressure cookers?

How do I certify LED lamps under CRS?



Ask about BIS certification, standards, or products...



AI guidance grounded on authoritative BIS records. Always verify Gazette mandates on bis.gov.in

Compliance Workspace

Suggested Standards & Quick Scenarios

Tap any common product below to instantly analyze its mandatory BIS standards, clauses, and laboratory readiness:



Laptop Computer

IS 13252



Electric Kettle

IS 302-2-15



Pressure Cooker

IS 2347



LED Lamp

IS 16102



Safety Helmet

IS 2925



Mobile Phone

IS 13252 & 16333



Ceiling Fan

IS 374



Packaged Water

IS 14543



AI Compliance Ready

Ask any question in the chat or pick a product chip to inspect verified clauses, roadmaps, and certification schemes.

NAVIGATION

 AI Assistant

 **Scan & Verify**

 BIS Services

Scan & Verify

Upload a product rating label or paste regulatory specifications to cross-reference authentic BIS licenses, detect counterfeit markings, and confirm quality control orders.

QUICK VERIFICATION SCENARIOS (CLICK TO TEST)

 Electric Kettle (ISI Mark)

 HP Laptop (CRS Scheme)

 Prestige Pressure Cooker

 Philips LED Lamp

  **Counterfeit Label (Fraud Demo)**



Drag & drop label image or document

or [browse files](#) · PNG, JPG, WebP up to 10MB

OR

Paste product rating plate or regulatory label text

e.g. Havells Kettle | IS 302 (Part 2/Sec 15) | CM/L-1234567 | 230V 1500W...

 **Analyze & Verify Label**

Verification Results

Authenticity check against indexed BIS Standards & Registries



No verification results yet

Upload an image or select a quick demo scenario on the left to verify authenticity against BIS databases.



AI Readiness

Compliance profile

68

/ 100

Complete a scan to improve your business score.

N



NAVIGATION



AI Assistant



Scan & Verify



BIS Services



AI Readiness
Compliance profile

68

/ 100

N

Complete a scan to improve your
business score.

Scan & Verify

Upload a product rating label or paste regulatory specifications to cross-reference authentic BIS licenses, detect counterfeit markings, and confirm quality control orders.

QUICK VERIFICATION SCENARIOS (CLICK TO TEST)



Electric Kettle (ISI Mark)



HP Laptop (CRS Scheme)



Prestige Pressure Cooker



Philips LED Lamp



Counterfeit Label (Fraud Demo)



Selected: fake.jpeg

or [browse files](#) · PNG, JPG, WebP up to 10MB

OR

Paste product rating plate or regulatory label text

e.g. Havells Kettle | IS 302 (Part 2/Sec 15) | CM/L-1234567 | 230V 1500W...



Analyze & Verify Label

Verification Results

Authenticity check against indexed BIS Standards & Registries

CONFLICT



COUNTERFEIT DETECTED



COUNTERFEIT WARNING: The licence 'R-93010030' is officially registered for 'Self-Ballasted LED Lamp' (IS 16102). The uploaded label forged this licence on a product displaying 'IS 13252 (PART 1)'.

EXTRACTED REGULATORY ENTITIES

Product Name

Laptop / Notebook Computer

Manufacturer / Brand

TechBrand

Indian Standard (IS)

IS 13252 (PART 1)

Licence / Registration

R-93010030

REGULATORY PATHWAY & ACTION

Official Portal:

services.bis.gov.in

NAVIGATION

 AI Assistant

 **Scan & Verify**

 BIS Services

Scan & Verify

Upload a product rating label or paste regulatory specifications to cross-reference authentic BIS licenses, detect counterfeit markings, and confirm quality control orders.

QUICK VERIFICATION SCENARIOS (CLICK TO TEST)

 Electric Kettle (ISI Mark)

 HP Laptop (CRS Scheme)

 Prestige Pressure Cooker

 Philips LED Lamp

  **Counterfeit Label (Fraud Demo)**



Selected: real1.jpeg

or [browse files](#) · PNG, JPG, WebP up to 10MB

OR

Paste product rating plate or regulatory label text

e.g. Havells Kettle | IS 302 (Part 2/Sec 15) | CM/L-1234567 | 230V 1500W...

 **Analyze & Verify Label**

Verification Results

Authenticity check against indexed BIS Standards & Registries

VERIFIED



Information Technology Equipment - Safety - Part 1: General Requirements

 Information Verified: The extracted standard 'IS 13252 (PART 1)' correctly applies to Laptop / Notebook Computer under Compulsory Registration Scheme (CRS Phase-I). The licence format conforms to official BIS guidelines.

EXTRACTED REGULATORY ENTITIES

Product Name
Laptop / Notebook Computer

Manufacturer / Brand
Lenovo

Indian Standard (IS)
IS 13252 (PART 1)

Licence / Registration
R-41013218

REGULATORY PATHWAY & ACTION

Scheme:

Compulsory Registration Scheme (CRS Phase-I)



AI Readiness
Compliance profile

68

/ 100

Complete a scan to improve your
business score.



BIS Sahayak

Compliance Assistant

NAVIGATION



AI Assistant



Scan & Verify



BIS Services



AI Readiness

Compliance profile

68

/ 100



Complete a scan to improve your
business score.

BIS Services

Explore Bureau of Indian Standards certification schemes and access the official portals to begin your application.



ISI Mark Certification

Mandatory certification scheme for products covered under compulsory registration. Ensures products conform to the relevant Indian Standard.

FOR: Manufacturers of domestic & industrial goods

[Visit Official Portal](#)



CRS Registration

Compulsory Registration Scheme for electronics and IT goods. Self-declaration of conformity backed by testing in BIS-recognized labs.

FOR: Electronics & IT product importers

[Visit Official Portal](#)



Hallmarking

Third-party assurance of the purity of gold and silver articles. Registration for jewellers and assaying & hallmarking centres.

FOR: Jewellers & bullion dealers

[Visit Official Portal](#)



Need help deciding?

Not sure which certification applies to your product? Let our AI assistant analyze your requirements and recommend the right path.

[Ask the AI Assistant →](#)